HyperSLR Digital Twin Framework

Coastal Flood Risk Assessment Report

Bangladesh | Vietnam | Maldives

Report date	2026-03-12 11:19:35 UTC
Baseline date	2022-12-31
System risk level	GREEN
LLM provider	offline-template
Framework version	HyperSLR Cell 38
Audience	National coastal adaptation planners

Country Risk Status

Country	Risk Level	Rise (mm)	Next Threshold	Years	Compound Prob
Bangladesh	GREEN	108.7	AMBER	7.4	0.0919
Vietnam	GREEN	0.0	AMBER	5.4	0.1000
Maldives	GREEN	0.0	AMBER	4.9	0.2000

Figure 1 — Cumulative Sea-Level Rise

Cumulative SLR anomaly (mm) above the 2022-12-31 baseline. Dashed/dotted horizontal lines indicate AMBER, RED, and CRITICAL risk thresholds. Grey vertical line marks the baseline date.

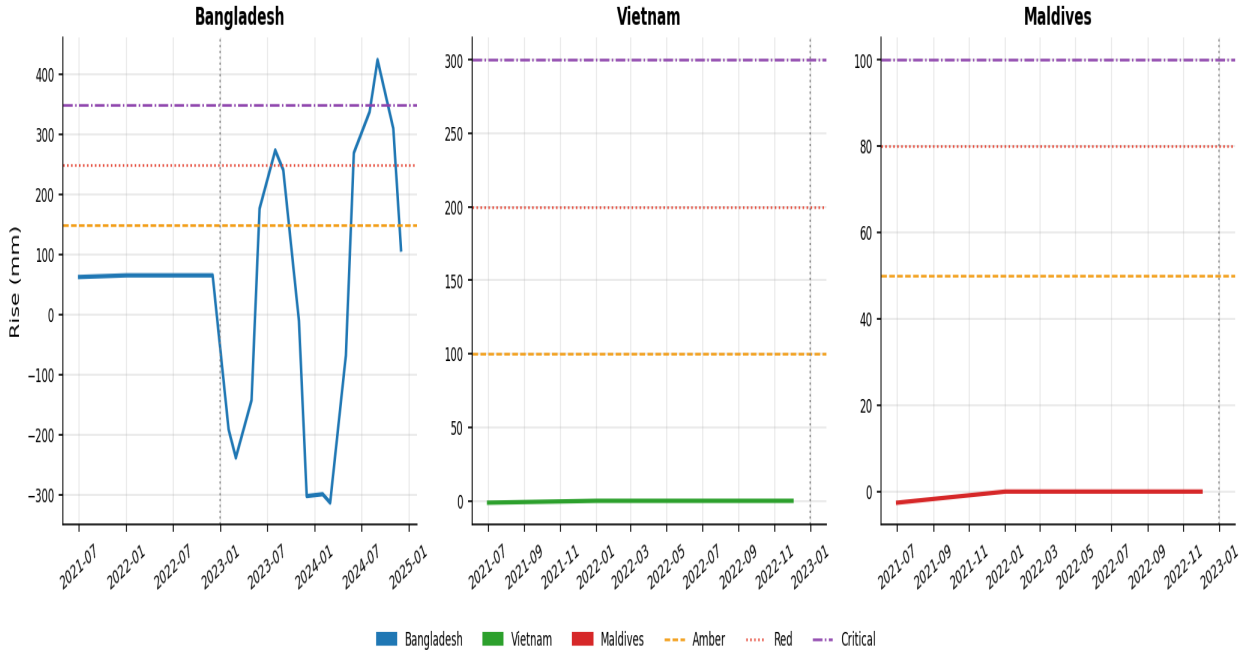


Figure 2 — Risk Gauges and Threshold Timeline

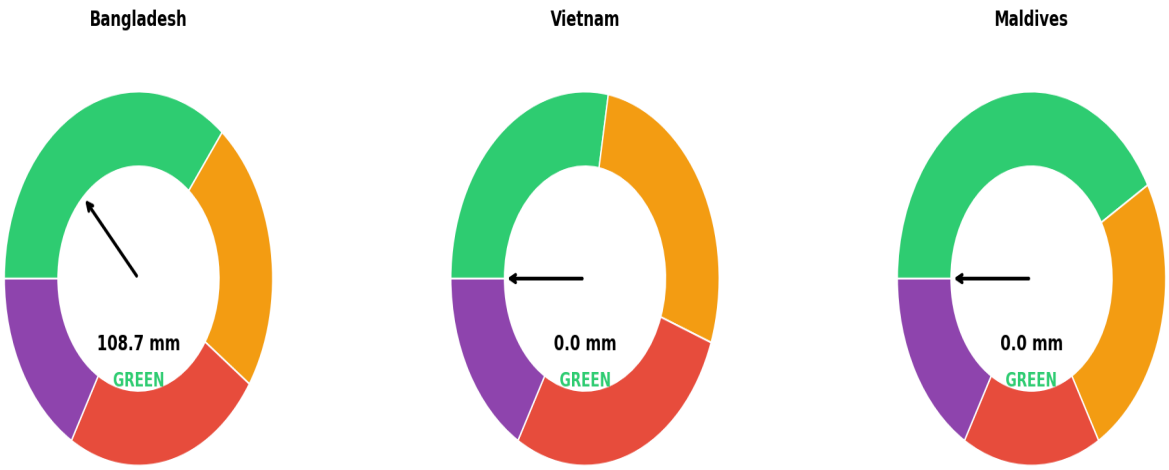


Figure 3 — Years to Risk Threshold (SSP2-4.5)

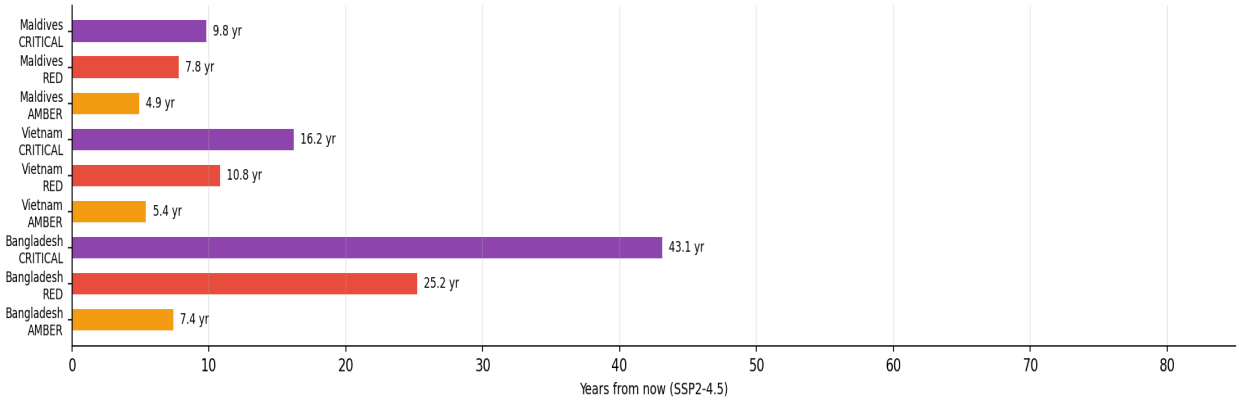


Figure 4 — Compound Event Probability Heatmap

Annual probability (%) of compound SLR exceedance + storm surge event, estimated via Gumbel copula (theta=2.0, SSP2-4.5). Red cell = exceedance probable in that decade.

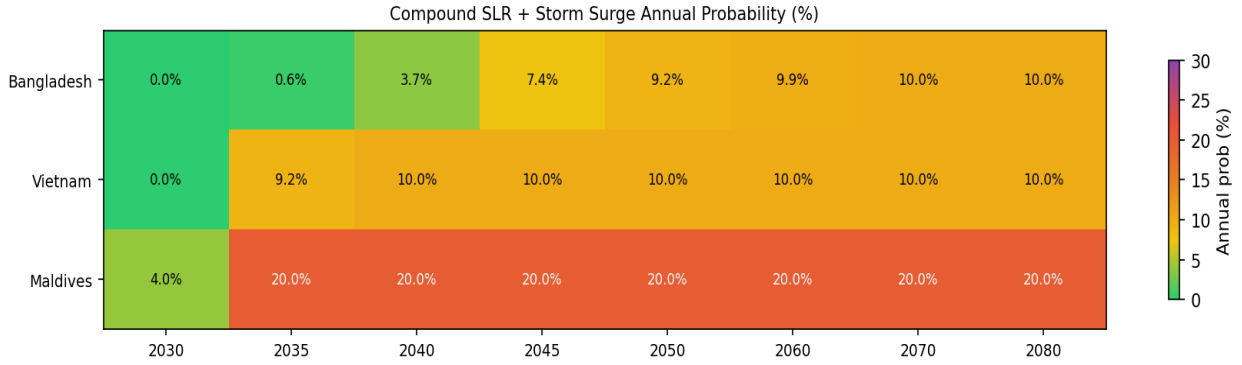


Figure 5 — Acceleration Z-Score (24-month rolling)

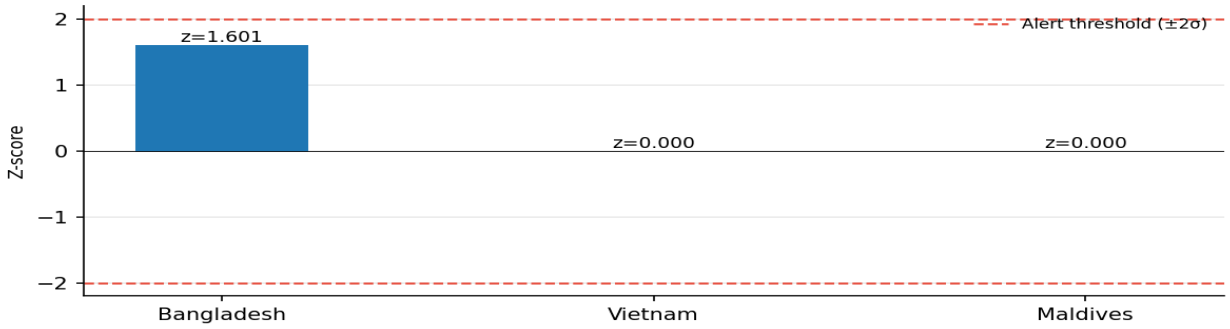


Table 1 — 5-Year Adaptation Measures

Country	Measure	Priority	Cost	Timeline
Bangladesh	Coastal Early Warning System Upgrade	HIGH	MEDIUM	2025-2026
Bangladesh	Mangrove and Coastal Ecosystem Restoration	HIGH	LOW	2025-2027
Bangladesh	Climate-Resilient Infrastructure Standards	HIGH	HIGH	2025-2028
Bangladesh	Managed Coastal Retreat Planning	MEDIUM	HIGH	2026-2030
Bangladesh	Subsidence Monitoring and Groundwater Governance	MEDIUM	MEDIUM	2025-2027
Vietnam	Coastal Early Warning System Upgrade	HIGH	MEDIUM	2025-2026
Vietnam	Mangrove and Coastal Ecosystem Restoration	HIGH	LOW	2025-2027
Vietnam	Climate-Resilient Infrastructure Standards	MEDIUM	HIGH	2025-2028
Vietnam	Managed Coastal Retreat Planning	MEDIUM	HIGH	2026-2030
Vietnam	Subsidence Monitoring and Groundwater Governance	HIGH	MEDIUM	2025-2027
Maldives	Coastal Early Warning System Upgrade	HIGH	MEDIUM	2025-2026
Maldives	Mangrove and Coastal Ecosystem Restoration	HIGH	LOW	2025-2027
Maldives	Climate-Resilient Infrastructure Standards	HIGH	HIGH	2025-2028
Maldives	Managed Coastal Retreat Planning	MEDIUM	HIGH	2026-2030
Maldives	Sea-Level Monitoring Station Densification	MEDIUM	MEDIUM	2025-2027
Regional	Bay of Bengal / Indian Ocean Regional SLR…	MEDIUM	LOW	2026-2028
Regional	Regional Climate Displacement Protocol	MEDIUM	MEDIUM	2027-2030

Adaptation Recommendations

Generated by: offline-template | Audience: National coastal adaptation planners

Bangladesh

5-Year Adaptation Measures

Coastal Early Warning System Upgrade

Timeline: 2025-2026 | Priority: HIGH | Cost Tier: MEDIUM

Bangladesh has recorded 108.7 mm cumulative SLR above the 2022 baseline with a compound flood probability of 0.0919 annually, requiring sub-seasonal warning lead times for low-lying coastal communities.

Actions:

- Deploy tide gauge telemetry at 15-minute resolution at all major estuarine entry points.
- Integrate CMEMS satellite altimetry into national flood forecasting models within 18 months.
- Establish SMS/community radio last-mile alert protocols reaching 90% of coastal ward populations.

Mangrove and Coastal Ecosystem Restoration

Timeline: 2025-2027 | Priority: HIGH | Cost Tier: LOW

Nature-based solutions can reduce wave energy by 50-70% and provide sediment accretion offsetting SLR at rates competitive with the 108.7 mm rise already observed.

Actions:

- Map and gazette 20% additional coastal buffer zones as protected mangrove corridors.
- Implement community-based mangrove nursery programmes targeting 5,000 ha restoration by 2027.
- Monitor ecosystem accretion rates against SLR annually using LiDAR elevation surveys.

Climate-Resilient Infrastructure Standards

Timeline: 2025-2028 | Priority: HIGH | Cost Tier: HIGH

With 7.4 years to the next risk threshold (AMBER), new coastal infrastructure must be designed for a 50-year SLR horizon under SSP2-4.5.

Actions:

- Revise national building codes to mandate freeboard allowances for SLR in all coastal construction permits.
- Conduct elevation surveys of critical infrastructure (roads, hospitals, power substations) within 2 years.
- Retire or elevate 100-year floodplain assets below the RED risk threshold elevation.

Managed Coastal Retreat Planning

Timeline: 2026-2030 | Priority: MEDIUM | Cost Tier: HIGH

CVI of 0.847 indicates high structural vulnerability. Proactive managed retreat planning reduces emergency displacement costs by up to 4:1 versus reactive post-disaster relocation.

Actions:

- Commission national coastal exposure assessment identifying settlements below RED threshold elevation.
- Develop voluntary relocation incentive schemes with land tenure guarantees for receiving areas.
- Pilot managed retreat in 3 high-exposure sub-districts by 2027 with full monitoring and evaluation.

Subsidence Monitoring and Groundwater Governance

Timeline: 2025-2027 | Priority: MEDIUM | Cost Tier: MEDIUM

VLM rate of -2.0 mm/yr significantly amplifies effective SLR beyond global mean, requiring dedicated monitoring and governance response.

Actions:

- Install continuous GNSS benchmarks at 10 coastal locations to monitor vertical land motion quarterly.
- Regulate groundwater extraction in coastal aquifer zones demonstrating >5 mm/yr subsidence rates.
- Publish annual VLM monitoring report informing local government land use decisions.

Policy Recommendations

National Coastal Adaptation Plan (NAP) Update

Lead Agency: Ministry of Environment / National Disaster Risk Reduction Agency | Horizon: short (1-2yr) | Urgency: IMMEDIATE

Existing NAPs predate the 108.7 mm observed rise and the compound probability data generated by this monitoring system; urgent revision is required.

Instruments:

- Mandatory 5-year NAP review cycle anchored to live SLR monitoring data from this HyperSLR system.
- Cross-ministerial coastal adaptation committee with defined decision triggers at AMBER, RED, CRITICAL thresholds.

Coastal Zone Land Use Moratorium Below AMBER Threshold Elevation

Lead Agency: Ministry of Land / Urban Planning Authority | Horizon: short (1-2yr) | Urgency: IMMEDIATE

With 7.4 years to the next threshold, permitting new permanent structures in the exposure zone locks in asset losses that exceed prevention costs by 3-8x.

Instruments:

- Issue ministerial directive restricting coastal construction permits below defined risk elevation contours.
- Establish risk-zone overlay in national GIS cadastre linked to HyperSLR threshold boundaries.

Climate Finance Mobilisation for Coastal Resilience

Lead Agency: Ministry of Finance / Climate Change Unit | Horizon: medium (3-5yr) | Urgency: NEAR-TERM

Green Climate Fund, Adaptation Fund, and World Bank PROBLUE resources are available but require pre-qualified project pipelines and monitoring frameworks to unlock.

Instruments:

- Develop a National Adaptation Finance Strategy identifying eligible instruments for each adaptation measure.
- Register this HyperSLR monitoring system as an MRV (Monitoring, Reporting, Verification) tool for GCF proposals.
- Engage NDC partnership for technical assistance in packaging bankable coastal resilience projects.

Coastal Fisheries and Livelihood Transition Programme

Lead Agency: Ministry of Fisheries / Social Affairs | Horizon: medium (3-5yr) | Urgency: STRATEGIC

Coastal livelihood diversification reduces population exposure dependency and improves voluntary relocation uptake.

Instruments:

- Aquaculture and mariculture development grants for coastal fishing communities.
- Social protection floor linked to coastal relocation to reduce resistance to managed retreat.

Vietnam

5-Year Adaptation Measures

Coastal Early Warning System Upgrade

Timeline: 2025-2026 | Priority: HIGH | Cost Tier: MEDIUM

Vietnam has recorded 0.0 mm cumulative SLR above the 2022 baseline with a compound flood probability of 0.1000 annually, requiring sub-seasonal warning lead times for low-lying coastal communities.

Actions:

- Deploy tide gauge telemetry at 15-minute resolution at all major estuarine entry points.
- Integrate CMEMS satellite altimetry into national flood forecasting models within 18 months.
- Establish SMS/community radio last-mile alert protocols reaching 90% of coastal ward populations.

Mangrove and Coastal Ecosystem Restoration

Timeline: 2025-2027 | Priority: HIGH | Cost Tier: LOW

Nature-based solutions can reduce wave energy by 50-70% and provide sediment accretion offsetting SLR at rates competitive with the 0.0 mm rise already observed.

Actions:

- Map and gazette 20% additional coastal buffer zones as protected mangrove corridors.
- Implement community-based mangrove nursery programmes targeting 5,000 ha restoration by 2027.
- Monitor ecosystem accretion rates against SLR annually using LiDAR elevation surveys.

Climate-Resilient Infrastructure Standards

Timeline: 2025-2028 | Priority: MEDIUM | Cost Tier: HIGH

With 5.4 years to the next risk threshold (AMBER), new coastal infrastructure must be designed for a 50-year SLR horizon under SSP2-4.5.

Actions:

- Revise national building codes to mandate freeboard allowances for SLR in all coastal construction permits.
- Conduct elevation surveys of critical infrastructure (roads, hospitals, power substations) within 2 years.
- Retire or elevate 100-year floodplain assets below the RED risk threshold elevation.

Managed Coastal Retreat Planning

Timeline: 2026-2030 | Priority: MEDIUM | Cost Tier: HIGH

CVI of 0.793 indicates high structural vulnerability. Proactive managed retreat planning reduces emergency displacement costs by up to 4:1 versus reactive post-disaster relocation.

Actions:

- Commission national coastal exposure assessment identifying settlements below RED threshold elevation.
- Develop voluntary relocation incentive schemes with land tenure guarantees for receiving areas.
- Pilot managed retreat in 3 high-exposure sub-districts by 2027 with full monitoring and evaluation.

Subsidence Monitoring and Groundwater Governance

Timeline: 2025-2027 | Priority: HIGH | Cost Tier: MEDIUM

VLM rate of -8.9 mm/yr significantly amplifies effective SLR beyond global mean, requiring dedicated monitoring and governance response.

Actions:

- Install continuous GNSS benchmarks at 10 coastal locations to monitor vertical land motion quarterly.
- Regulate groundwater extraction in coastal aquifer zones demonstrating >5 mm/yr subsidence rates.
- Publish annual VLM monitoring report informing local government land use decisions.

Policy Recommendations

National Coastal Adaptation Plan (NAP) Update

Lead Agency: Ministry of Environment / National Disaster Risk Reduction Agency | Horizon: short (1-2yr) | Urgency: IMMEDIATE

Existing NAPs predate the 0.0 mm observed rise and the compound probability data generated by this monitoring system; urgent revision is required.

Instruments:

- Mandatory 5-year NAP review cycle anchored to live SLR monitoring data from this HyperSLR system.
- Cross-ministerial coastal adaptation committee with defined decision triggers at AMBER, RED, CRITICAL thresholds.

Coastal Zone Land Use Moratorium Below AMBER Threshold Elevation

Lead Agency: Ministry of Land / Urban Planning Authority | Horizon: short (1-2yr) | Urgency: IMMEDIATE

With 5.4 years to the next threshold, permitting new permanent structures in the exposure zone locks in asset losses that exceed prevention costs by 3-8x.

Instruments:

- Issue ministerial directive restricting coastal construction permits below defined risk elevation contours.
- Establish risk-zone overlay in national GIS cadastre linked to HyperSLR threshold boundaries.

Climate Finance Mobilisation for Coastal Resilience

Lead Agency: Ministry of Finance / Climate Change Unit | Horizon: medium (3-5yr) | Urgency: NEAR-TERM

Green Climate Fund, Adaptation Fund, and World Bank PROBLUE resources are available but require pre-qualified project pipelines and monitoring frameworks to unlock.

Instruments:

- Develop a National Adaptation Finance Strategy identifying eligible instruments for each adaptation measure.
- Register this HyperSLR monitoring system as an MRV (Monitoring, Reporting, Verification) tool for GCF proposals.
- Engage NDC partnership for technical assistance in packaging bankable coastal resilience projects.

Groundwater Extraction Regulation in Coastal Aquifer Zones

Lead Agency: Ministry of Water Resources / Geological Survey | Horizon: medium (3-5yr) | Urgency: NEAR-TERM

VLM of -8.9 mm/yr is substantially driven by groundwater abstraction in deltaic aquifers; regulatory intervention can reduce effective SLR by 2-4 mm/yr.

Instruments:

• Groundwater extraction licensing with volumetric caps in designated coastal subsidence risk zones.
• Annual aquifer level monitoring with automatic permit suspension if drawdown exceeds thresholds.

Maldives

5-Year Adaptation Measures

Coastal Early Warning System Upgrade

Timeline: 2025-2026 | Priority: HIGH | Cost Tier: MEDIUM

Maldives has recorded 0.0 mm cumulative SLR above the 2022 baseline with a compound flood probability of 0.2000 annually, requiring sub-seasonal warning lead times for low-lying coastal communities.

Actions:

• Deploy tide gauge telemetry at 15-minute resolution at all major estuarine entry points.
• Integrate CMEMS satellite altimetry into national flood forecasting models within 18 months.
• Establish SMS/community radio last-mile alert protocols reaching 90% of coastal ward populations.

Mangrove and Coastal Ecosystem Restoration

Timeline: 2025-2027 | Priority: HIGH | Cost Tier: LOW

Nature-based solutions can reduce wave energy by 50-70% and provide sediment accretion offsetting SLR at rates competitive with the 0.0 mm rise already observed.

Actions:

• Map and gazette 20% additional coastal buffer zones as protected mangrove corridors.
• Implement community-based mangrove nursery programmes targeting 5,000 ha restoration by 2027.
• Monitor ecosystem accretion rates against SLR annually using LiDAR elevation surveys.

Climate-Resilient Infrastructure Standards

Timeline: 2025-2028 | Priority: HIGH | Cost Tier: HIGH

With 4.9 years to the next risk threshold (AMBER), new coastal infrastructure must be designed for a 50-year SLR horizon under SSP2-4.5.

Actions:

• Revise national building codes to mandate freeboard allowances for SLR in all coastal construction permits.
• Conduct elevation surveys of critical infrastructure (roads, hospitals, power substations) within 2 years.
• Retire or elevate 100-year floodplain assets below the RED risk threshold elevation.

Managed Coastal Retreat Planning

Timeline: 2026-2030 | Priority: MEDIUM | Cost Tier: HIGH

CVI of 0.921 indicates high structural vulnerability. Proactive managed retreat planning reduces emergency displacement costs by up to 4:1 versus reactive post-disaster relocation.

Actions:

• Commission national coastal exposure assessment identifying settlements below RED threshold elevation.
• Develop voluntary relocation incentive schemes with land tenure guarantees for receiving areas.
• Pilot managed retreat in 3 high-exposure sub-districts by 2027 with full monitoring and evaluation.

Sea-Level Monitoring Station Densification

Timeline: 2025-2027 | Priority: MEDIUM | Cost Tier: MEDIUM

Continuous in-situ monitoring underpins all risk threshold detection and early warning systems.

Actions:

• Commission 3 additional PSMSL-affiliated tide gauges in under-monitored coastal sectors.
• Establish data-sharing agreement with regional oceanographic agencies for real-time altimetry.
• Publish open-access coastal monitoring dashboard updated monthly.

Policy Recommendations

National Coastal Adaptation Plan (NAP) Update

Lead Agency: Ministry of Environment / National Disaster Risk Reduction Agency | Horizon: short (1-2yr) | Urgency: IMMEDIATE

Existing NAPs predate the 0.0 mm observed rise and the compound probability data generated by this monitoring system; urgent revision is required.

Instruments:

- Mandatory 5-year NAP review cycle anchored to live SLR monitoring data from this HyperSLR system.
- Cross-ministerial coastal adaptation committee with defined decision triggers at AMBER, RED, CRITICAL thresholds.

Coastal Zone Land Use Moratorium Below AMBER Threshold Elevation

Lead Agency: Ministry of Land / Urban Planning Authority | Horizon: short (1-2yr) | Urgency: IMMEDIATE

With 4.9 years to the next threshold, permitting new permanent structures in the exposure zone locks in asset losses that exceed prevention costs by 3-8x.

Instruments:

- Issue ministerial directive restricting coastal construction permits below defined risk elevation contours.
- Establish risk-zone overlay in national GIS cadastre linked to HyperSLR threshold boundaries.

Climate Finance Mobilisation for Coastal Resilience

Lead Agency: Ministry of Finance / Climate Change Unit | Horizon: medium (3-5yr) | Urgency: NEAR-TERM

Green Climate Fund, Adaptation Fund, and World Bank PROBLUE resources are available but require pre-qualified project pipelines and monitoring frameworks to unlock.

Instruments:

- Develop a National Adaptation Finance Strategy identifying eligible instruments for each adaptation measure.
- Register this HyperSLR monitoring system as an MRV (Monitoring, Reporting, Verification) tool for GCF proposals.
- Engage NDC partnership for technical assistance in packaging bankable coastal resilience projects.

Coastal Fisheries and Livelihood Transition Programme

Lead Agency: Ministry of Fisheries / Social Affairs | Horizon: medium (3-5yr) | Urgency: STRATEGIC

Coastal livelihood diversification reduces population exposure dependency and improves voluntary relocation uptake.

Instruments:

- Aquaculture and mariculture development grants for coastal fishing communities.
- Social protection floor linked to coastal relocation to reduce resistance to managed retreat.

Regional / Cross-Border

5-Year Adaptation Measures

Bay of Bengal / Indian Ocean Regional SLR Monitoring Consortium

Timeline: 2026-2028 | Priority: MEDIUM | Cost Tier: LOW

Bangladesh, Vietnam, and the Maldives share oceanographic drivers (ENSO, IOD, thermosteric expansion) that are most efficiently monitored and modelled at regional scale.

Actions:

- Formalise a tripartite data-sharing agreement linking national tide gauge networks under IOGOOS.
- Establish a shared CMEMS sub-regional altimetry processing pipeline updated monthly.
- Joint capacity-building programme for national hydrological services in PINN-based SLR forecasting.

Regional Climate Displacement Protocol

Timeline: 2027-2030 | Priority: MEDIUM | Cost Tier: MEDIUM

Maldivian and low-lying Bangladesh coastal populations face displacement scenarios that require pre-agreed regional frameworks before crisis conditions force ad-hoc responses.

Actions:

- Commission a regional climate displacement needs assessment under SAARC / IORA auspices.
- Negotiate bilateral labour mobility agreements enabling climate-linked migration pathways.
- Establish a regional climate displacement fund with initial capitalisation from Green Climate Fund.

Policy Recommendations

UNFCCC National Adaptation Plan (NAP) Alignment and Joint Reporting

Lead Agency: UNFCCC National Focal Points / SAARC Environment Ministers | Horizon: medium (3-5yr) | Urgency: NEAR-TERM

Coordinated NAP submissions under the UNFCCC NAP process strengthen collective bargaining for adaptation finance and technical support.

Instruments:

- Joint submission to UNFCCC NAP Technical Working Group presenting this HyperSLR system as shared MRV infrastructure.
- Coordinated Loss and Damage claims documentation using common HyperSLR baseline and risk metrics.

Regional Catastrophe Risk Insurance Pool

Lead Agency: Regional Development Banks / Finance Ministries | Horizon: long (5-10yr) | Urgency: STRATEGIC

Individual national insurance markets lack the risk diversification to offer affordable coastal flood cover; a regional pool following the CCRIF model is viable.

Instruments:

- Feasibility study for a Bay of Bengal / Indian Ocean Catastrophe Risk Pool modelled on CCRIF SPC.
- Parametric insurance product design using HyperSLR risk thresholds as trigger conditions.

Monitoring Triggers

Bangladesh cumulative rise reaches AMBER threshold — Threshold: 150 mm above 2022-12-31 baseline

Response: Convene emergency NAP review; activate Tier 2 coastal protection budget; issue construction moratorium in RED exposure zone.

Vietnam or Maldives cumulative rise reaches 25 mm above baseline — Threshold: 25 mm (50% of AMBER threshold)

Response: Initiate managed retreat site identification; accelerate early warning system deployment; trigger GCF project preparation funding.

Acceleration z-score exceeds 2.0 for any country — Threshold: z > 2.0 on 24-month rolling trend

Response: Revise SSP2-4.5 time-to-threshold projections upward; brief cabinet; engage IPCC WGI national focal point for expert assessment.

Annual compound event probability exceeds 5% — Threshold: compound_prob_annual > 0.05

Response: Declare coastal climate emergency; activate national disaster risk reduction budget supplementary allocation.

Critical Uncertainties

- SSP2-4.5 rate assumptions may underestimate acceleration if West Antarctic Ice Sheet instability progresses faster than IPCC AR6 median projections.
- VLM rates for Bangladesh and Vietnam are based on sparse GNSS networks; true subsidence may exceed modelled rates in intensively pumped agricultural zones.
- Compound event probability estimates assume stationarity of storm surge climatology; climate-driven intensification of tropical cyclones could increase return period frequencies.
- Political will and institutional capacity for managed retreat remain the primary implementation bottleneck, not technical or financial constraints.